DS

GATE CS / IT 2027

Data Structures & Algorithms

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

GATEFlow Preparation Platform

DS Data Structures & Algorithms

GATE CS/IT

Sorting Algorithms

Algorithm	Best	Average	Worst	Space	Stable
Bubble	O(n)	O(n2)	O(n2)	O(1)	Yes
Selection	O(n2)	O(n2)	O(n2)	O(1)	No
Insertion	O(n)	O(n2)	O(n2)	O(1)	Yes
Merge	O(nlogn)	O(nlogn)	O(nlogn)	O(n)	Yes
Quick	O(nlogn)	O(nlogn)	O(n2)	O(logn)	No
Heap	O(nlogn)	O(nlogn)	O(nlogn)	O(1)	No
Counting	O(n+k)	O(n+k)	O(n+k)	O(k)	Yes

Tree Data Structures

BST & AVL

- BST: inorder = sorted, ops O(h)
- AVL: |h_left - h_right| <= 1, ops O(logn)
- AVL rotations: LL, RR, LR, RL
- Red-Black Tree: also O(logn), used in C++ STL

Heap & B-Tree

- Max-Heap: parent >= children; Min-Heap: parent <= children
- Build heap: O(n); Insert/Delete: O(logn)
- B-Tree of order m: max m children, min ceil(m/2) children
- B+ Tree: all data in leaves, internal nodes for routing
- B+ Tree supports range queries efficiently

Graph Algorithms

Traversal & Shortest Path

- BFS: O(V+E), uses queue, shortest path in unweighted graph
- DFS: O(V+E), used for topological sort, SCC, cycle detection
- Dijkstra: O((V+E)logV), no negative edges
- Bellman-Ford: O(VE), handles negative edges, detects cycles
- Floyd-Warshall: O(V3), all-pairs shortest path

MST & Connectivity

- Kruskal: sort edges + Union-Find = O(E logE)
- Prim: O(E logV) with min-heap, grow tree greedily
- Both give same MST cost, MST has exactly V-1 edges
- SCC: Kosaraju (2 DFS) or Tarjan algorithm
- Topological sort only for DAGs (DFS or Kahn's BFS)

Dynamic Programming

Classic DP Problems

- 0/1 Knapsack: dp[i][w] = max(no include, include item i)
- LCS(s,t): dp[i][j] = dp[i-1][j-1]+1 if s[i]==t[j], else max(dp[i-1][j], dp[i][j-1])
- Edit Distance: min(insert, delete, replace) operations
- Matrix Chain: minimize multiplications, O(n^3)
- Coin Change: min coins to make amount = dp[amount]